

PERSONAL DETAILS

Name	Varun Anbalagan
Address	S-2, Star Apartments, Villivakkam, Chennai - 600049, Tamil Nadu, India
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Date of Birth	15.01.2004
Nationality	Indian
Marital Status	Single

EDUCATION

Dates	Details
08/2021 – Present	Bachelor of Technology in Computer Science and Engineering with Specialization in Artificial Intelligence and Machine Learning Vellore Institute of Technology (VIT), Chennai, India CGPA: 8.75 Thesis/Capstone: Fairness-Driven Legal Judgment Prediction with GNNs Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Machine Learning, Deep Learning, Graph Neural Networks, Theory of Computation, Discrete Mathematics, Linear Algebra.
06/2018 – 07/2021	All India Senior School Certificate Examination (AISSCE) - Class XII SBOA School and Junior College, Chennai, India Stream: Science (Computer Science, Mathematics, Physics, Chemistry) Result: 94.8%

Dates	Details
06/2014 – 05/2018	All India Secondary School Examination (AISSE) - Class X SBOA School and Junior College, Chennai, India Result: 90.2% Awards: First Place in Academic Proficiency (2014, 2015, 2016, 2017).

ACADEMIC PROJECTS & RESEARCH EXPERIENCE

Dates	Details
12/2024 – 04/2025	FairGavel – Fairness-Driven Legal Judgment Prediction with GNNs <i>Capstone Project, VIT Chennai</i> - Led the research and development of a bias-mitigating legal judgment prediction system utilizing Graph Neural Networks (GNNs) to address ethical AI constraints in the judiciary. - Engineered a graph construction pipeline employing BERT for semantic sentence node embedding and Named Entity Recognition (NER) for legal entity linking. - Architected a training framework integrating a GraphSAGE encoder with adversarial debiasing via Gradient Reversal Layer (GRL) and attention pooling mechanisms.
06/2024 – 12/2024	Legal Text Summarizer using Graph Convolutional Networks (GCNs) <i>Research Project, VIT Chennai</i> - Developed an extractive summarization tool for dense legal corpora by modeling documents as heterogeneous graph structures. - Tokenized legal texts using BERT embeddings; constructed graphs representing sentences and entities with multi-type edges to capture inter-clause dependencies. - Implemented a multi-layer GCN to aggregate contextual features, fused with a Transformer encoder-decoder for coherent summary generation. - Demonstrated improved interpretability and ROUGE scores compared to standard LSTM-based baselines.

Dates	Details
08/2023 – 12/2024	Non-Intrusive Stress Prediction via Sleep Position Analysis & IR Imaging <i>Research Project, VIT Chennai</i> • Designed a privacy-preserving stress estimation system using low-light infrared imaging and pose estimation techniques. • Curated a custom dataset using mannequin-based IR simulation; utilized YOLOv8 for the detection and normalization of 17 discrete body landmarks. • Extracted geometric features (Euclidean distances, angular symmetry, spread) and fused them into a Landmark Fusion model. • Deployed a regression model capable of real-time stress quantification based on sleep posture variations.
12/2023 – 04/2024	TextToTune – Lyric-Based Melody Generation <i>Course Project, VIT Chennai</i> • Investigated the intersection of NLP and creative AI by building a system to generate melodies from textual lyrics. • Processed a 12,000-song MIDI-lyrics dataset, extracting syllable-note alignments for supervised training. • Implemented a Conditional LSTM-GAN (Generative Adversarial Network), where both the generator and discriminator were conditioned on lyric embeddings. • Successfully generated coherent melodic sequences that maintained rhythmic alignment with input lyrical phrasing.
12/2023 – 04/2024	Real-Time Multimodal Sign Language Detection <i>IoT & AI Project, VIT Chennai</i> • Engineered a low-latency American Sign Language (ASL) translation system on edge devices (Raspberry Pi). • Optimized MediaPipe-based hand tracking models for ARM architecture to ensure real-time inference. • Established a wireless video streaming pipeline using ESP32 for real-time remote capture and processing.

Dates	Details
04/2023 – 07/2023	Animal Intrusion Detection with Deep Learning <i>Project, VIT Chennai</i> • Developed a computer vision system to detect and classify wildlife intrusion in agricultural zones. • Curated and annotated a dataset of 2,000 animal images; trained a CNN-based classification model. • Deployed the model via a Streamlit web application, integrating SMS alert functionality (Twilio API) for real-time user notification.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Dates	Details
05/2022 – Present	Lead Game Developer & Team Manager <i>VIT Chennai</i> • Directed a 5-member development team in creating a 3D immersive game using Unreal Engine 5 and C++. • Architected core gameplay mechanics and physics interactions using Blueprints and C++ scripting. • Managed the full development lifecycle from concept to deployment; showcased the final build at "Vibrance," the university's cultural festival.
02/2023 – 03/2023	Committee Lead – Food and Logistics <i>Vibrance Cultural Fest, VIT Chennai</i> • Managed logistics for a 3-day campus-wide event attended by over [Number] students. • Coordinated with [Number] external vendors, negotiated contracts, and oversaw budget allocation. • Streamlined food court operations during peak hours, ensuring zero downtime during major concert events.

Dates	Details
01/2020 – 04/2021	Indie Game Developer <i>Independent Pursuit</i> - Developed "Cyber Combat," a 2D horizontal scroller using Python (Pygame). - Designed level architecture, enemy AI behavior trees, and save-state serialization systems.

CERTIFICATIONS

Dates	Details
2023	AWS Certified Cloud Practitioner Amazon Web Services (AWS)
2023	Google Cloud Digital Leader Google Cloud
2023	Entrepreneurship NPTEL
2022	Ethical Hacking NPTEL
2021	C, Advanced CPP, PHP and MySQL, Java, Python Spoken Tutorial

TECHNICAL SKILLS

Category	Details
Programming Languages	Python, C++, Java, C#, SQL, JavaScript

Category	Details
Machine Learning & AI	PyTorch, TensorFlow, Keras, Scikit-learn, Graph Neural Networks (GNNs), GANs, NLP (BERT, Transformers), Computer Vision (YOLO, MediaPipe)
Development Tools	Git, Docker, Unity, Unreal Engine, Arduino IDE, VS Code
Cloud & Databases	AWS (Cloud Practitioner), Google Cloud Platform, MySQL, MongoDB
Hardware	Raspberry Pi, ESP32, Arduino

LANGUAGES

Language	Proficiency Level
English	C1 (Proficient User) – TOEFL(106), Medium of Instruction in University
Tamil	Native Speaker
Hindi	C2 (Proficient User)
French	A2 (Basic User)